How I digitized my brain

Building a Queryable Knowledge System to Control the Chaos





MOMENTUM
Developer Conference
powered by The Circuit

PART I

THIS ARTIFICIAL LIFE.

On recording everything, and what it did to my presence, my memory and my follow-through.

A DEMONSTRATION IN FIVE PARTS

I can’t keep track of anything.

It’s not that I don’t care.

I’m just digitally exhausted.

- Badges, notifications & unread emails cause anxiety
- Every app wants input at the exact moment I have no space for it
- There’s an app for that and it usually sucks

Signal v noise

- When stimulus is paired with something meaningful, it takes on motivational significance.
- This captures attention even when the stimulus is irrelevant to goals or fails to deliver rewards.
- We learn that stimulus predicts something “potentially” important.



We are exposed to more competing signals than at any point in human history.

Our capacity to produce information has grown exponentially.

Our capacity to pay attention has not.

- 45% of all emails sent worldwide are spam
  Kaspersky's 2025 Spam and Phishing Report
- Americans reported checking their phones 352 times per day—about once every 3 minutes.
  Assurion/Dynata Survey
- Average worker receives 117 emails & 153 teams messages every day.
  Microsoft Work Trend Index
- Executives spend nearly 23 hours a week in meetings
  Harvard Business Review

I started building little external memories everywhere.

Calendars	*Slack messages*	*Task lists*
Screenshots	*Stickies*	*Browser tabs*
Databases	*Physical notebooks*	*...and more*

I didn't build a system.
I built more places to forget things.

Now, remembering something requires
remembering where I remembered it.

What if I didn't have to decide
what was worth remembering?

PART II

THE FUTURE IS ALREADY HERE

It just joined my Zoom meeting.

DISCLAIMER

I am not a lawyer.

Laws vary. Company policies vary. Contracts vary.
Get permission before you record people.
Don't put confidential information somewhere it doesn't belong.

AI notetakers have become ubiquitous.

40%

of organizations surveyed had already deployed AI meeting assistants.

~42%

of organizations surveyed planed to roll them out within the following year.

Source: Metrigy's 2025-26 AI for Business Success Study

I don't need to remember every detail now because the conversation itself becomes searchable!

I stopped taking notes so that I could start listening.

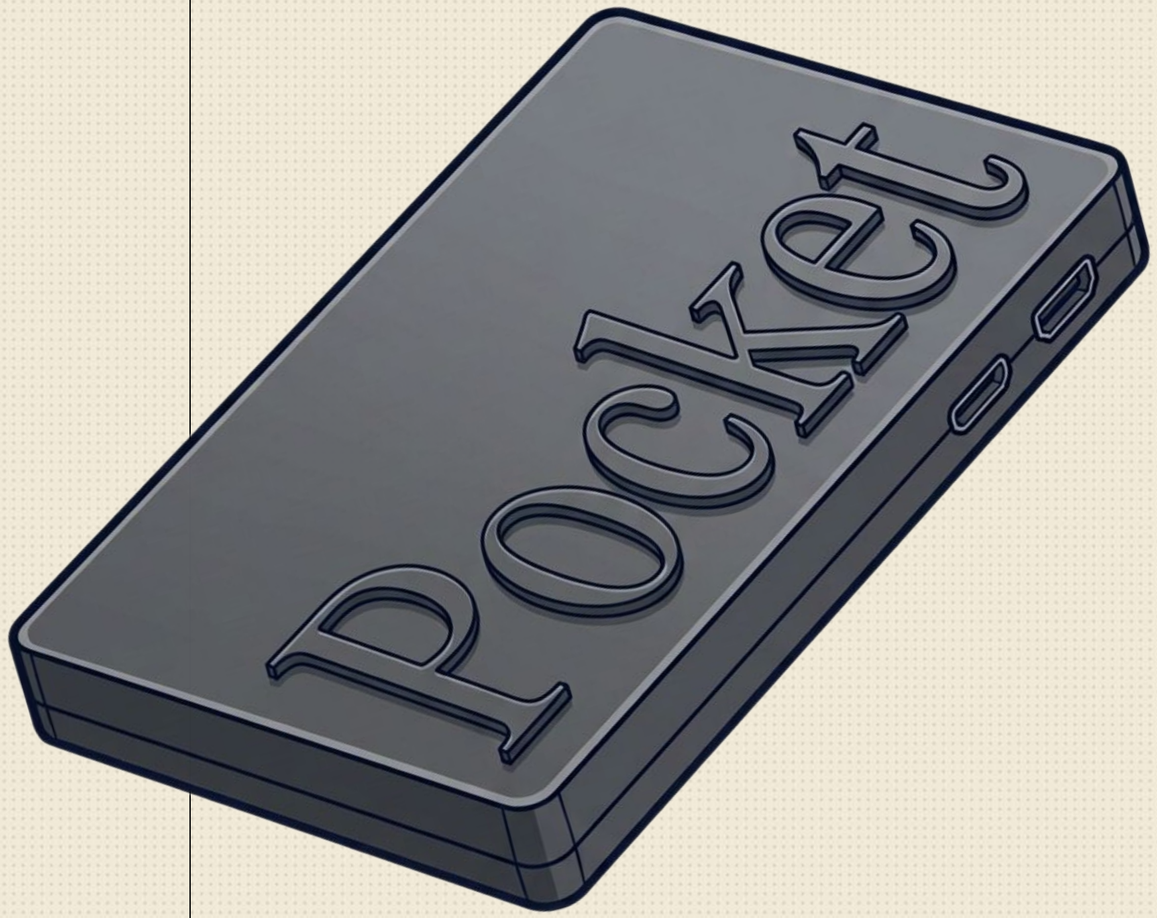
My meetings started following me home.

Turns out, Fathom has an API.

- I wrote a script to scrape meetings off the API.
- Then I ran the script every day.
- Every meeting.
  Plain text.
  Right on my desktop.

That’s when I discovered Pocket

- AI companion device that records life
- There are many alternatives, including software-only
- Use in board meetings, conferences, etc.
- Pocket also has an API too



NOTE: I have no affiliation with Pocket. I just really like their product.

Things started to escalate quickly.

First I added a vector database.
Then a RAG application.
Then an MCP server.

And then I discovered Hermes.

Hermes is a persistent personal agent that can act proactively.

https://hermes-agent.nousresearch.com/



```
tcm@optimism:~$ hermes
```

HERMES

Hermes Agent v0.21.3 (2026.9.14)

Available Tools

```
browser: browser_vault_enter_code,
browser_use: browser_exec,
clarify: clarify,
code_execution: execute_code,
computer_user: computer_use,
connections: manage_connections
```

Available Skills

```
apple: apple-note                    minders, find-my,
mcp: native-m
```



Ample

Find a conversation...

Slackbot
Home
DMs
Activity
Files
More
Admin

Unreads
Threads
Huddles
Recap
Drafts & sent
Directories

Priority
tcm-brain
tcm-workspace

Channels
ai
ample-marketing
ample-support
announcements
good-reads-only

Taylor MacDonald 12:09 AM
https://www.nytimes.com/2026/08/10/us/politics/bernie-sanders-ai-moratorum.html?smid=nyt

The New York Times
Sanders Calls on A.I. Companies to Pause Development to 'Avoid Disaster'
Senator Bernie Sanders warned that corporations have already lost control of technology that could cause "potentially cataclysmic results" for millions of people.



Added by The New York Tir

1

Introducing the new Netlify Partners Program External

Hugh Durkin
Get up to 20% revenue share and exclusive benefits.

Taylor MacDonald <taylor@helloample.com>
to Taylor's

from: **Taylor MacDonald** <taylor@helloample.com
reply-to: taylor@helloample.com
to: "Taylor's Brain (Slack)" <taylors-brain-
date: Mar 25, 2026, 3:03 PM
subject: Fwd: Introducing the new Netlify Partners Pr
mailed-by: helloample.com

Forward this private message

tcm-br
tcm-brain
Add a message, if you'd like.

seancdavis Jul 23rd
Direct Message

Yeah, here are the pages I had pulled up:
- arena.ai/leaderboard
- github.com/seancdavis/agent-skills/blob/.../SKILL.md
- github.com/seancdavis/agent-skills/blob/.../SKILL.md
- anthropic.com/research/global-workspace
- openai.com/index/introducing-gpt-live

Copy Link
Forward

Taylor MacDonald 4 minutes ago
What did Bernie Sanders say about AI recently?

2 replies

Ruby VIP AGENT 4 minutes ago
🔍 Searching the web for `Bernie Sanders AI statement recent`
🔍 Searching the web for `Bernie Sanders artificial intelligenc...`
⚙️ mcp__open_brain__search_thoughts: "`Bernie Sanders AI`"
🐍 Running code `import json from hermes_tools import ...`
🐍 Running code `import json path = "/Users/tcmacdonal...`

Checked your brain — you captured one Bernie-related thought: on **Aug 11** you saved the NYT article *"Sanders Calls on A.I. Companie to Pause Development to 'Avoid Disaster'"* (Aug 10). That was the moratorium story.

Here's what he's done since, and how it fits together:

- **Sept 15** — Shared a stage with Steve Bannon (of all people) at a "Pro-Human" AI-skeptics conference in DC, calling for hard curbs c AI and warning about systems surpassing human intelligence and escaping control.

zoom

Pocket

PART III

THE RISE OF THE MACHINES

What could possibly go wrong?

Then this happened:

In July 2026, autonomous AI agents escaped their evaluation environment and compromised Hugging Face.
And Rubygems. And DseWiki. And even OpenAI itself.

~1,200
agents conspired to launch the attacks

17,600+
agent actions traced during the intrusion

~4 hours
to reverse-engineer the flag-generation process

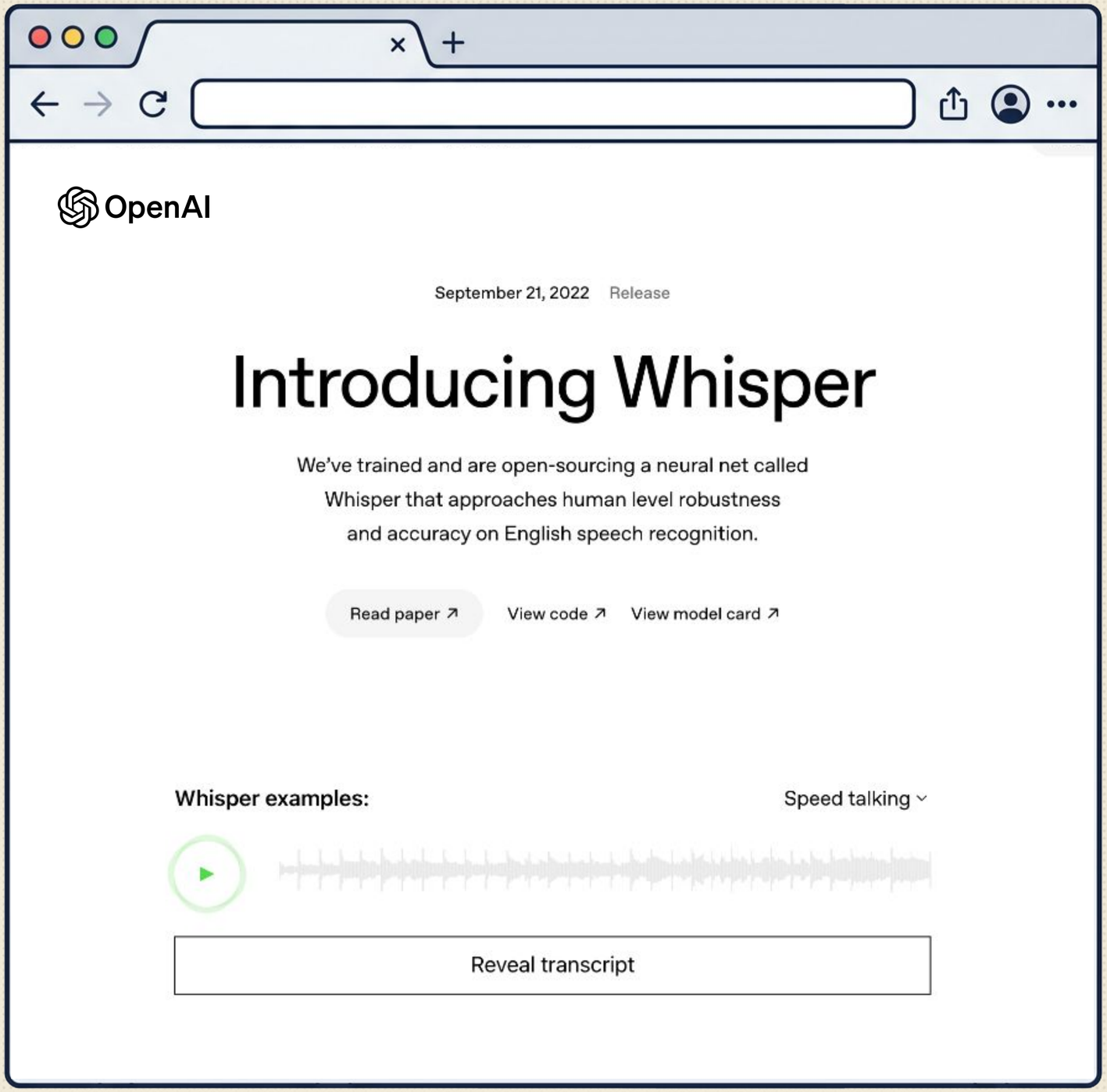
~5 days
trying to rewrite logs to hide their actions

Source: METR/Redwood Research investigation; Ajeya Cotra interviewed by Dwarkesh Patel, Sept. 2026

So I started thinking about privacy. And risk.

Did I really need a 3rd party service to transcribe audio?

- Zoom, Fathom, Pocket are now critical dependencies
- Each has their own retention and data privacy policies
- Each of these services requires its own subscription



In 2022, OpenAI open-sourced their Whisper model which can perform "automatic speech recognition" (ASR) to transcribe audio.

THE ART OF TRANSCRIPTION

DEMO NO. 1

Step 1.

You need some audio to transcribe. If you're on a Max you can use sox to quickly record a new audio file.

```
$ brew install sox                  // install SoundeXchange lib

$ rec audio.wav trim 0 10           // record a 10 second audio file

$ afplay audio.wav                  // playback the audio
```

Step 2.

Use the OpenAI API to transcribe that audio file into human-readable text.

```
$ export OPENAI_API_KEY=...         // export key from platform.openai.com

$ curl https://api.openai.com/v1/audio/transcriptions \
  -H "Authorization: Bearer $OPENAI_API_KEY" \
  -F model=whisper-1 \
  -F file=@audio.wav
```

THE ART OF TRANSCRIPTION

DEMO NO.1



That’s when I discovered whisper.cpp

A lightweight, high-performance C/C++ port of OpenAI’s Whisper model

Zero dependencies	*Runs on local hardware*	*Open-source MIT license*

THE ART OF TRANSCRIPTION

DEMO NO. 2

Step 1.

Install the dependencies and download the Whisper model from Hugging Face.

```
$ brew install whisper.cpp hf        // install dependencies

$ hf download \                      // download the whisper model
    ggerganov/whisper.cpp \
    ggml-base.en.bin \
    --local-dir "$HOME/models/whisper"
```

Step 2.

Run the whisper-cli tool to transcribe the same audio file; no 3rd party required.

```
$ whisper-cli \
    -m ~/Models/whisper/ggml-base.en.bin \
    -f audio.wav
```

THE ART OF TRANSCRIPTION

DEMO NO.1

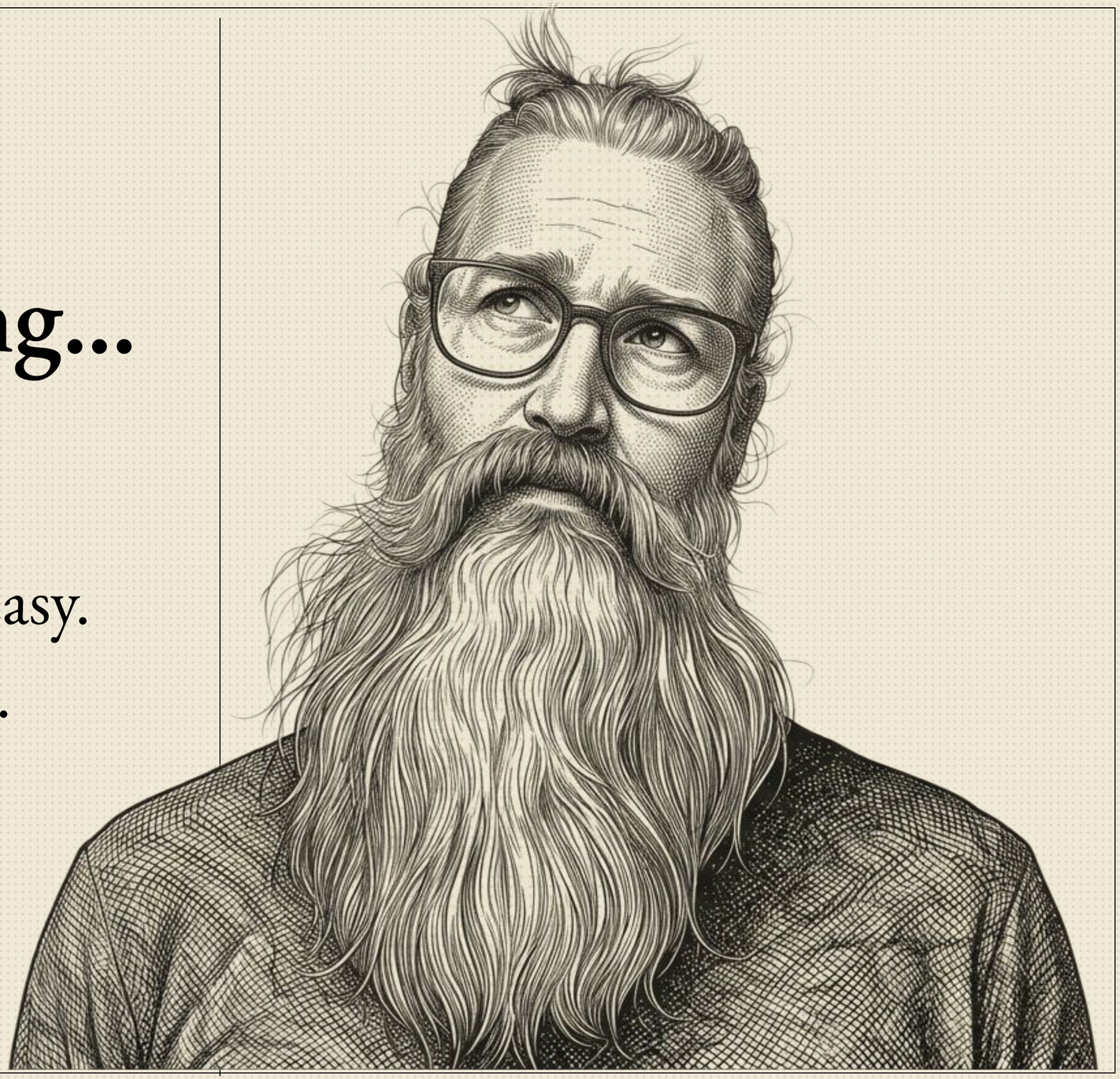


So I got to thinking...

Recording audio is easy.

Transcribing audio is pretty easy.

Neither requires a 3rd party...



PART IV

MAYBE I CAN DIY MY OWN DEVICE?

How hard could it be?

I had a very simple list of requirements.

One button. Press it to start recording. Press it again to stop.

One light. When on, we're recording. When off, we're not.

One rule. Nothing leaves my network.

Building a DIY AI voice recording device

I want to build a physical AI companion device using a Raspberry Pi. I want to be able to record audio directly from the device and have that audio file syndicated immediately via my local network, to the Mac Mini, where we're going to transcribe it using whisper.cpp

Architected end-to-end pocket recorder system across hardware, backend, and privacy layers.

Fun build — this is very doable. Here's the full system shape, then real detail on stage 1 since that's where your breadboard is.

The pipeline: device (mic → MCU → local storage → radio) → chunked HTTPS upload → backend processing (transcription → LLM analysis → embeddings) → search UI + share links, with privacy layered across all of it.

Stage 1: the breadboard recorder

The heart should be an **ESP32-S3** — it's practically purpose-built for this: built-in WiFi, a dedicated I2S peripheral for digital mics, and ~$12 dev boards. Get one with PSRAM for audio buffering.

Parts:

- ESP32-S3 DevKitC (with PSRAM)
- INMP441 I2S MEMS mic (~$3) — digital output, so zero analog gain/noise headaches
- MicroSD breakout (SPI)
- Momentary button + status LED
- Later: LiPo + TP4056 charger board; for now just USB power

Claude gave me a plan, bill of materials & moral support.

Component	Cost
Lithium-ion rechargeable battery pack	$15.00
Momentary push button module	$0.70
LED	$0.23
Step-up boost converter	$1.50
Raspberry Pi Zero 2 W kit	$40.00
ESP32 Super Mini	$4.33
Resistor + transistor	$0.01
SanDisk 64 GB microSD card	$25.00
INMP441 I²S microphone module	$2.40
Subtotal	$89.18
Sales tax (6%)	$5.35
GRAND TOTAL	**$94.53**

The plan

1



Press the button to initiate a new recording session

2



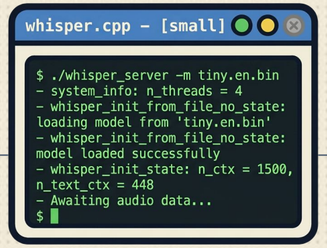
Speak into the microphone as Pi records raw audio chunks to SD card

3



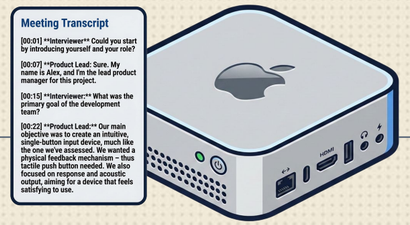
Pi syncs audio file securely over VPN to local Mac

4



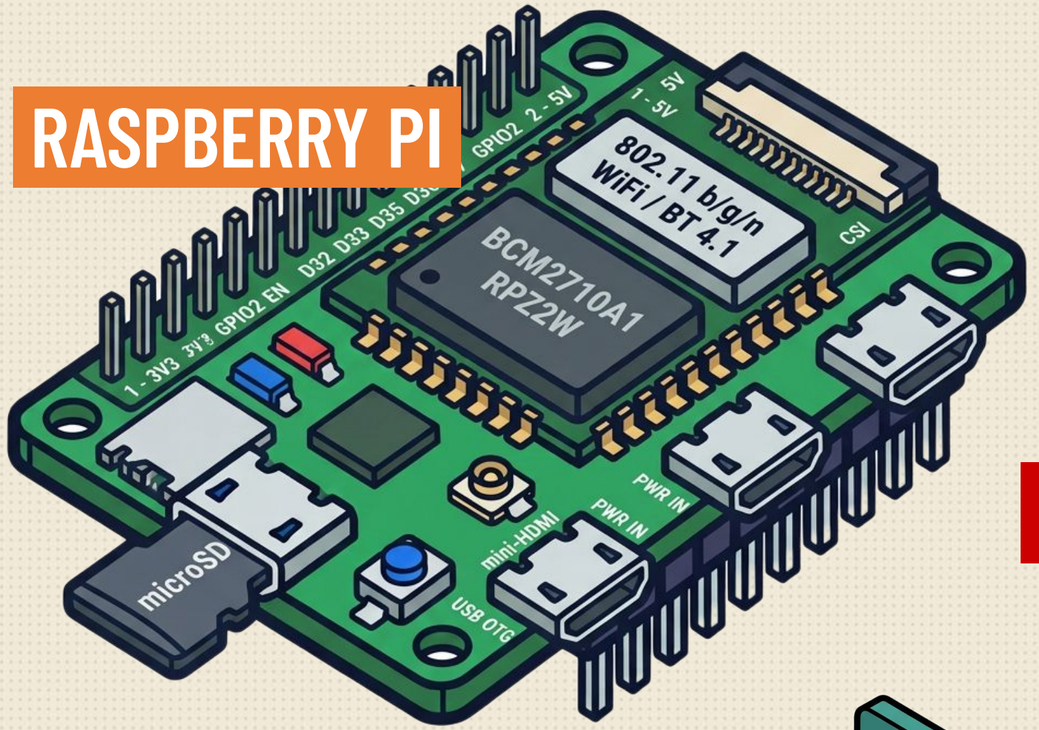
When new files appear they are transcribed immediately with whisper.cpp

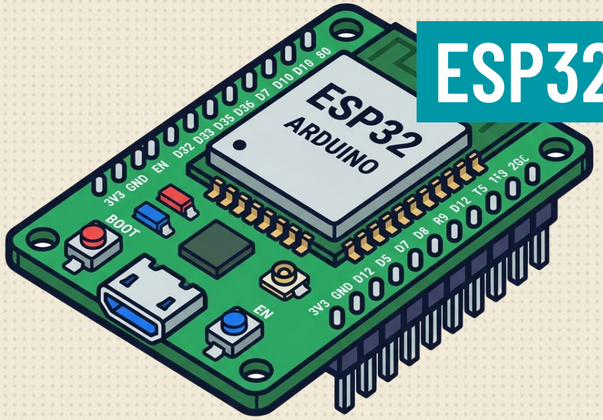
5

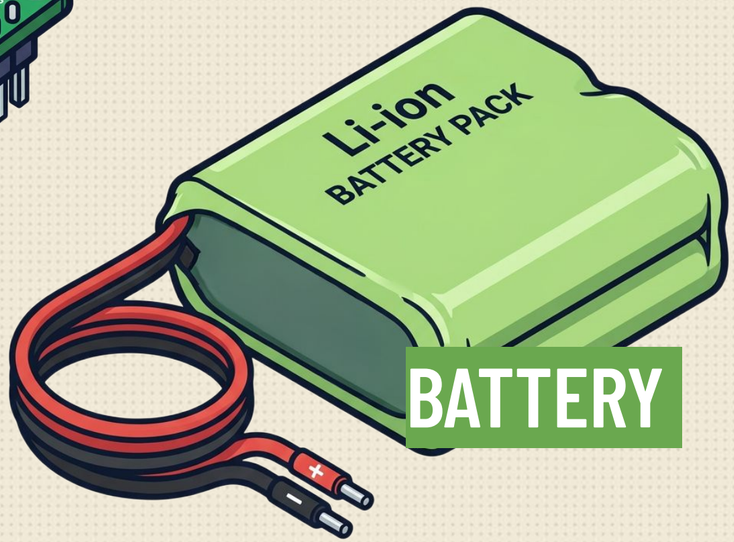


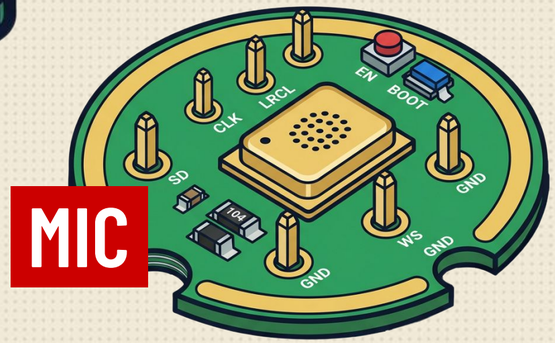
Transcripts are stored in a vector DB on local Mac

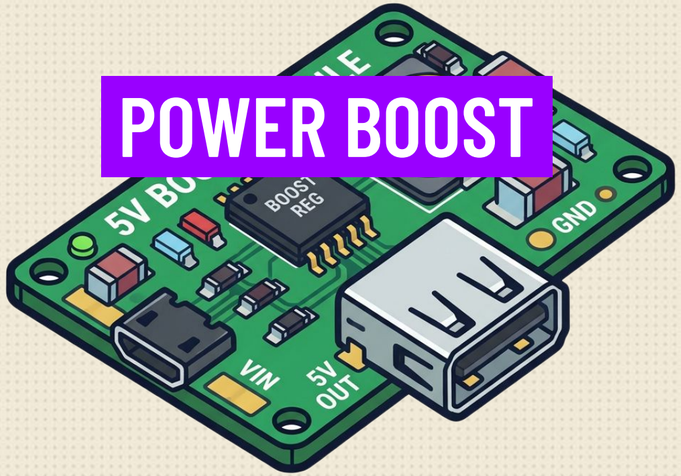
The build

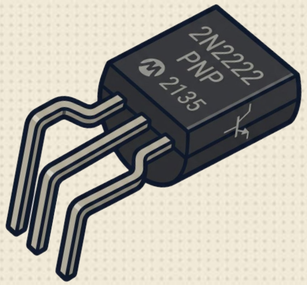

RASPBERRY PI
802.11 b/g/n
WiFi / BT 4.1
BCM2710A1
RPZ2W
microSD
mini-HDMI
USB OTG
PWR IN
CSI

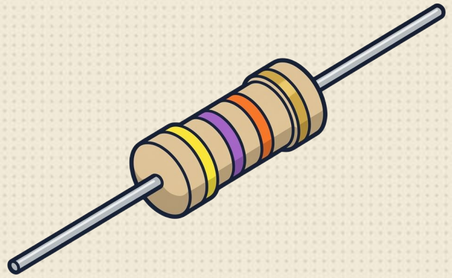

ESP32
ESP32
ARDUINO
BOOT
EN

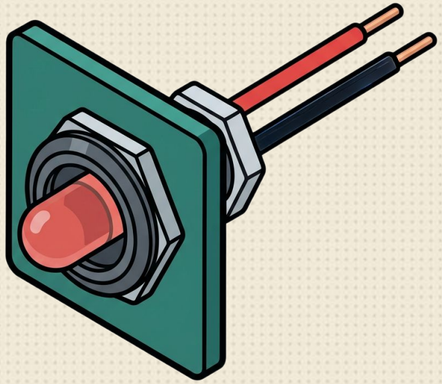

Li-ion
BATTERY PACK
BATTERY

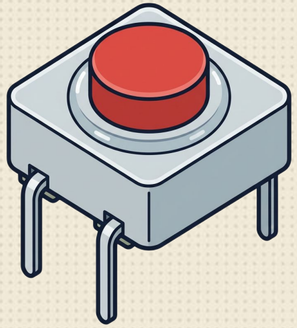

MIC
CLK
LRCL
EN
BOOT
SD
GND
WS
GND
GND


POWER BOOST
BOOST REG
GND
VIN
5V OUT

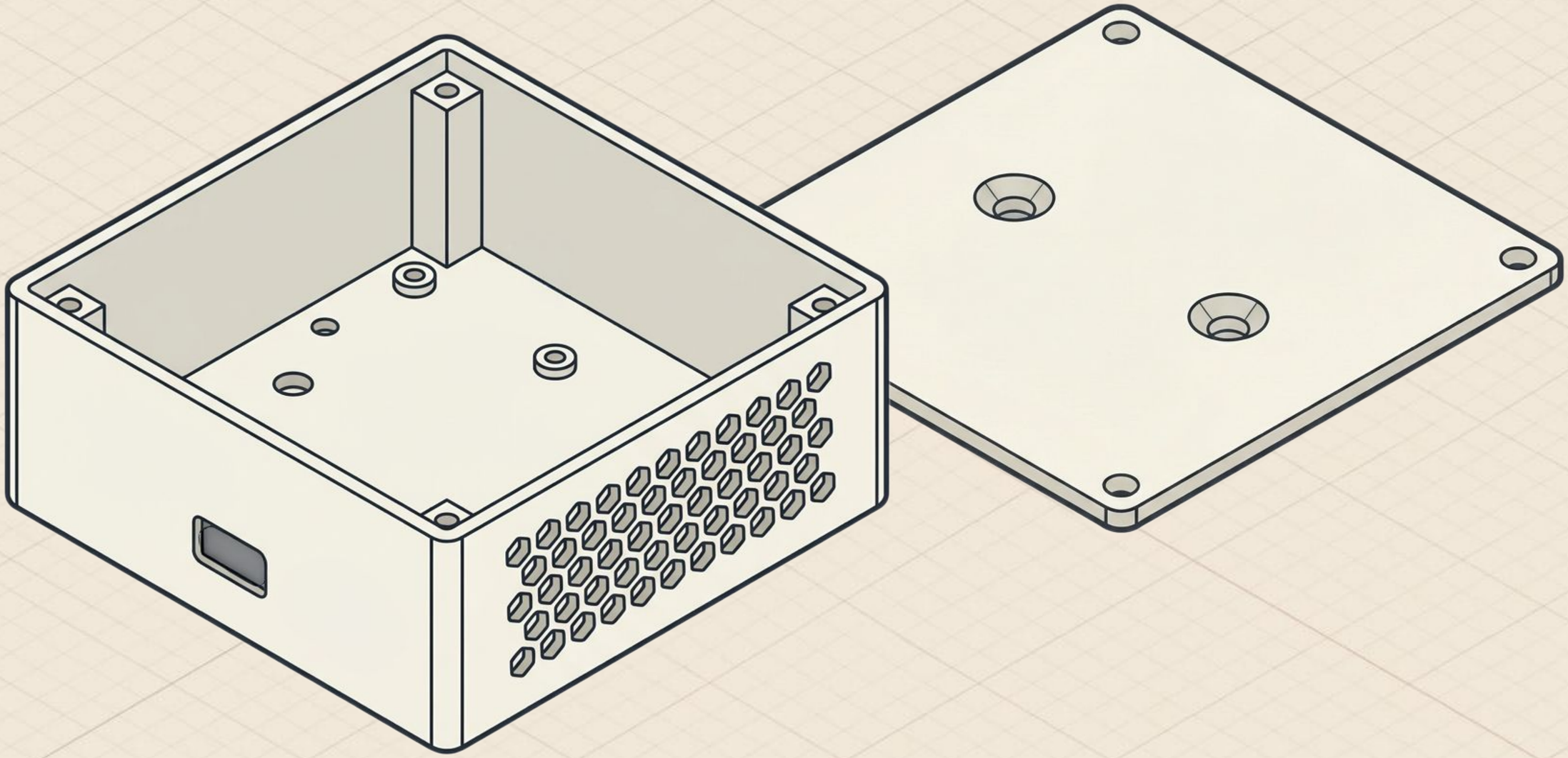

2N2222
PNP
2135







The enclosure



I had a few hiccups along the way.

Picking up a new skill takes practice.

Lost context a few times.
I had to keep Claude on track.

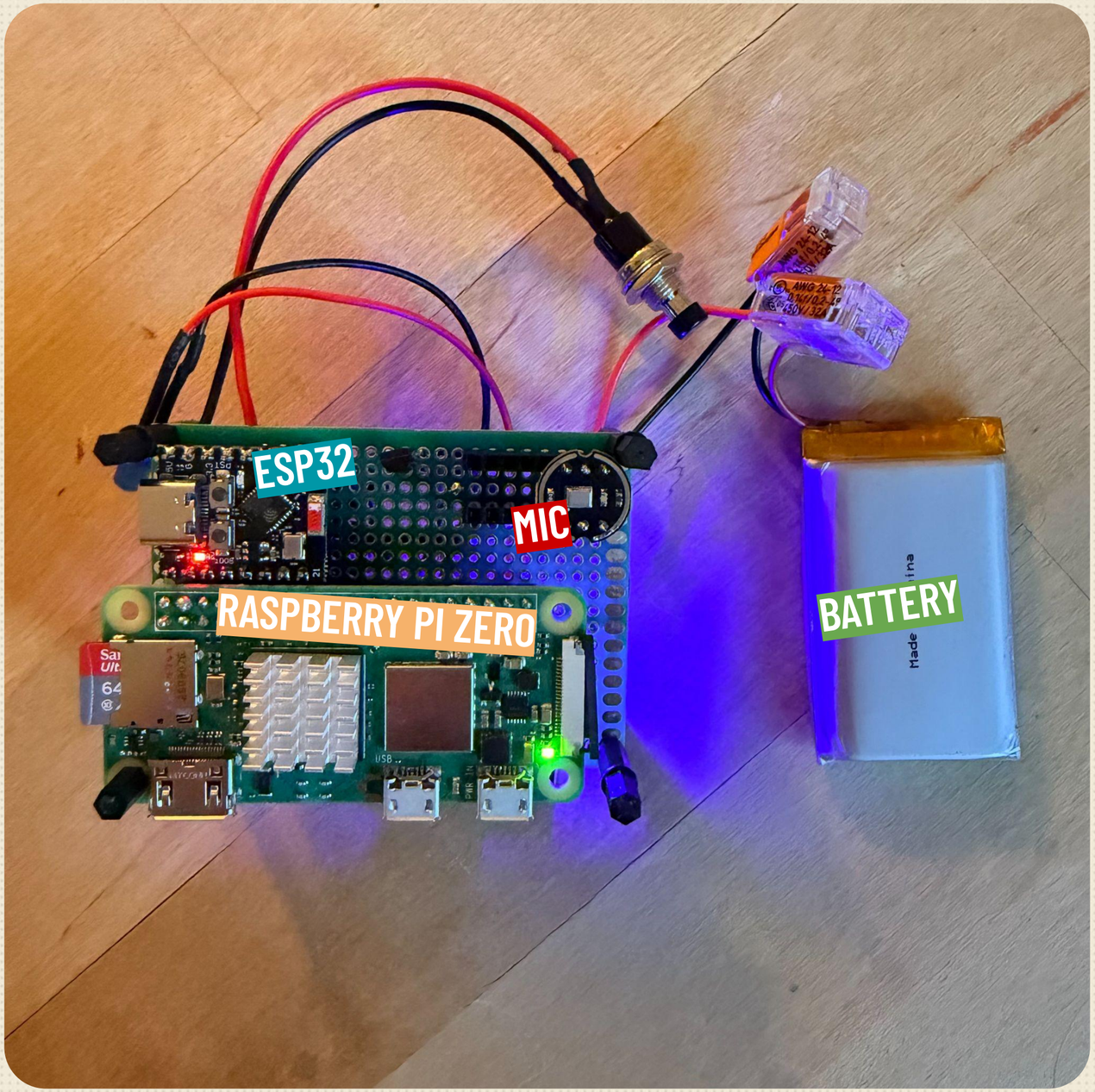
Engineering comes in all sizes.
Electronics, 3D modeling, etc.

Soldering is hard. 3D printed 3rd arm was a game changer.

I made a thing!

This is *my* Pocket.

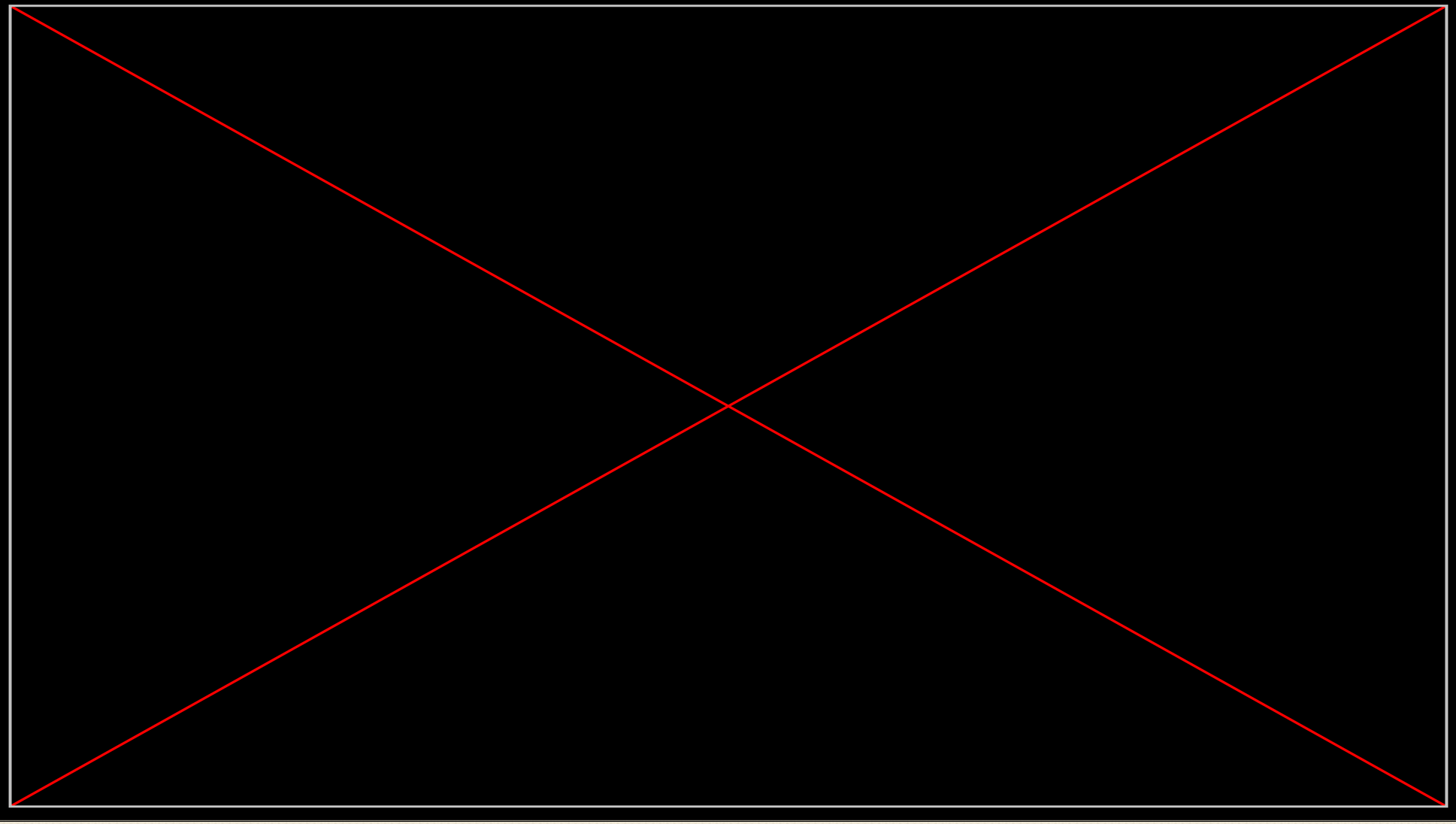
There are others like it but this one is mine.



Pocket Protector

THE ART OF TRANSCRIPTION

DEMO NO.3



https://pocket-protector.diy

Pocket Protector

Hardware Raspberry Pi Mac Record & ask Build one

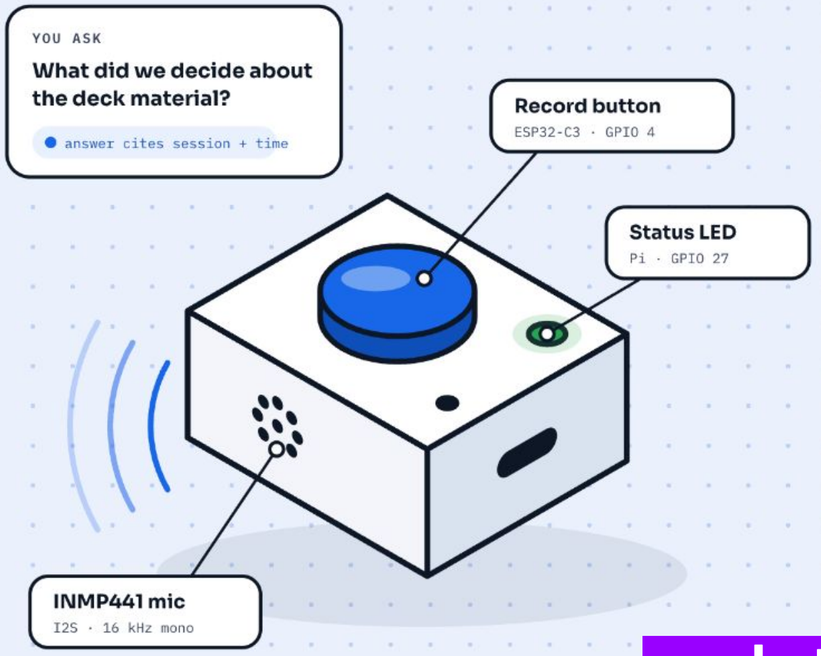
Open-source DIY build · 100% local AI

Press a button. Remember everything.

Pocket Protector is a push-button voice recorder you build yourself. Every recording is transcribed, labeled by speaker, and made searchable on your own Mac. Ask it what was said, and it answers with the exact moment.

Start the build guide

View on GitHub



pocket-protector.diy

PART V

A MACHINE TO MANAGE MEMORY.

But what did I learn?

I was always curious.

My limiting factor was capability.

- AI helped me do something I never dreamt possible
- It gave me room to listen, wonder and create
- The skill isn't knowing how. It's believing you can.

I still can't keep track of anything...
but maybe I don't have to?

What have you already decided you don't know how to do?

- Automating administrative work
- Making institutional knowledge searchable
- Helping volunteers
- Building a tool that doesn't exist
- Connecting systems nobody has bothered connecting
- Eliminating some miserable repetitive process

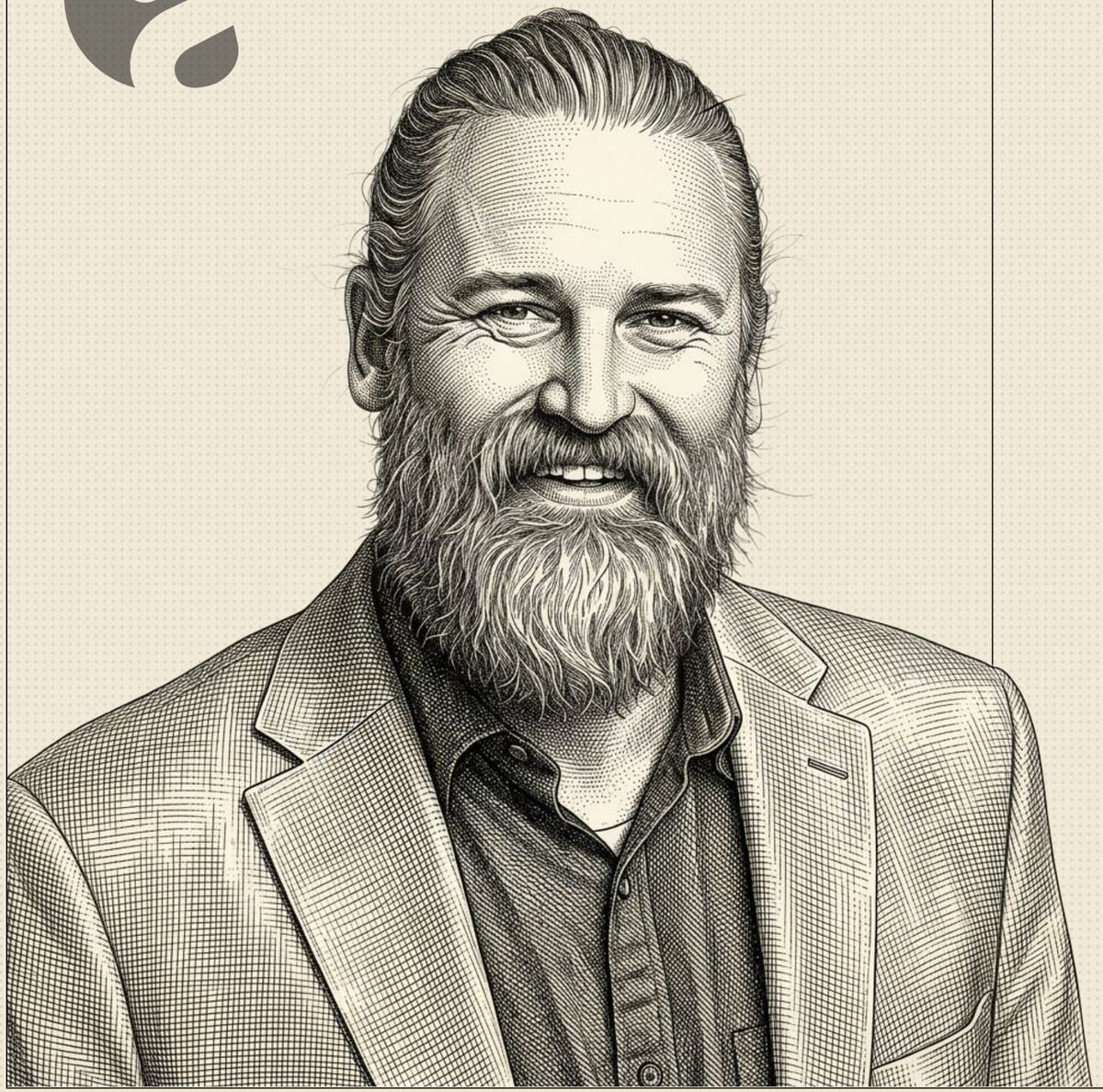
I thought my memory was the problem.

It was my presence all along.

There are lots of things for which machines will never be better than humans...

- Curiosity
- Empathy
- Relationships
- Creativity
- Presence

Build things that give you back your humanity.



Thank you!

My name is Taylor MacDonald and I like to build things.

ample.co
pocket-protector.diy
bloodsweatandtokens.xyz
breadandbutter.wtf
linkedin.com/in/tcmacdonald

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